

Zhuo Chen

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Summary

I'm a software engineer who enjoys working on systems that need to be clear, dependable, and easy to operate at scale. I've worked on internal platforms, analytics engines, and cloud services at Meta and Oracle, where I spent most of my time with Python, Node.js, Java, and C#/.NET. I like solving the unglamorous problems—cleaning up backend logic, aligning APIs, improving observability, and building small AI-assisted features that actually make daily engineering work easier. My goal on any team is to make systems understandable and give other engineers a better experience using them.

Skills

AI & Machine Learning

OpenAI API, LangChain & LangGraph, vector search basics, embedding pipelines, retrieval augmentation, FastAPI inference servers, prompt routing, tool-calling flows, evaluation/test harnesses

Backend & APIs

Python 3.10–3.12 (FastAPI 0.110, Flask 2.x), Node.js 16–20 (Express 4, NestJS 9), Java 11–17 (Spring Boot 2.7–3.2), C#/.NET 6 & EF Core, Async I/O, REST/GraphQL, caching (Redis), AWS SDK, OAuth2/JWT, blue-green deploys, microservice decomposition, API versioning

Frontend & UI

React (17–18), TypeScript, Redux Toolkit, Storybook components, basic Next.js use, WebSockets, Jest & testing patterns, UI automation familiarity (Selenium), cross-team design reviews, Figma handoff experience

Data Engineering & Streaming

Apache Kafka (2.8–3.5), outbox pattern, exactly-once semantics, Airflow 2.4–2.9 (DAG validation, SLA monitoring), PySpark basics, Snowflake, PostgreSQL 13–16, MySQL 5.7/8, schema enforcement, Parquet processing

Cloud & DevOps

AWS: Lambda, S3, CloudFront, API Gateway, ECS/Fargate, ECR, DynamoDB basics, Cognito, IAM, CDK v2

Azure: Service Bus, App Service

Kubernetes (EKS), Docker, Terraform, GitHub Actions, Jenkins, Helm, rollout strategies

Observability & Reliability

OpenTelemetry tracing, structured logging, debugging workflows, Prometheus/Grafana dashboards, effective alarm tuning, SLO thinking, incident response, runbooks and cleanup documentation

Languages & Foundations

Python, Java, C#, TypeScript/JavaScript, SQL, Bash, Linux, algorithms, data structures, distributed system basics, system design

Soft Skills

Direct communication, mentoring, design discussions, cross-team alignment, documentation habits, clarity in ambiguity

Experience

Software Engineer — Meta (Full-time)

Seattle, WA • Dec 2021 – Present

Teams: Instagram / FB Analytics Platform (AI Engineer), Meta Observability Platform (RTC Observability)

Tech: Python, React/TypeScript, GraphQL, Spark/Presto, Scuba, OpenTelemetry, AWS-internal equivalents, internal RPC frameworks

Instagram / FB Analytics Platform — AI Engineer

Built AI-powered diagnostic tools and modernized platform APIs supporting Instagram and Facebook analytics.

AI-Assisted Engineering & Operational Intelligence

- Built an LLM-driven diagnostic assistant combining OpenAI models with internal retrieval indexes (logs, traces, config, past incidents).
- Improved on-call triage efficiency by automatically surfacing the most relevant operational signals.
- Implemented guardrails: ranking heuristics, structured prompts, fallback logic, and hallucination mitigation for production-safety.

Analytics API & Full-Stack Platform Modernization

- Refactored Python + GraphQL APIs powering high-traffic metrics dashboards (React/TypeScript frontends), reducing p95 latency by **30–40%**.
- Unified several legacy API behaviors behind a schema evolution framework to remove breaking changes.
- Implemented idempotent request/refresh patterns that stabilized bursty dashboard traffic during global analytics refreshes.
- Collaborated with frontend teams to adopt shared UI data contracts and improve type safety across backend–frontend boundaries.

Meta Observability Platform — Software Engineer (RTC Observability)

Built and improved observability systems used company-wide for debugging and visibility.

Tracing, Logging & Observability Foundations

- Rolled out OTEL tracing for Python-based analytics services, improving cross-service request visibility.
- Built structured logging schemas and ingestion classifiers that cut debugging cycles from hours of Scuba queries to minutes.
- Designed Grafana dashboards grounded in SLOs (freshness, latency, dependency health) for incident response.

Developer Tooling & Reliability

- Standardized log correlation and trace propagation patterns across analytics and infra teams.
- Developed runtime “issue detectors” that caught trace gaps, payload drift, and backend/ML integration issues early.
- Drove adoption of observability best practices across ML, analytics, and infra teams.

Senior Software Engineer — Oracle (Full-time)

Broomfield, CO • Jul 2017 – Dec 2021

Teams: OCI Identity & Access Management (IAM), SaaS Billing & Revenue Management (BRM)

Tech: Java, Spring Boot, Node.js, .NET Core, Oracle DB, Kafka, Airflow, Docker, Kubernetes, Prometheus/Grafana

OCI Identity & Access Management (IAM)

Identity, entitlement, and authorization systems used across Oracle Cloud.

- Built a unified entitlement/feature-gating API used by **20+ Oracle SaaS product teams**, replacing fragmented authorization rules.
 - Defined API versioning and contract testing that reduced cross-team regressions and supported gradual rollout.
 - Migrated monolithic IAM behaviors into domain-aligned microservices using Spring Boot and .NET Core.
 - Designed internal debugging tools to help support teams inspect complex license state transitions without code access.
 - Introduced structured-event logging standards that improved auditability and compliance visibility.
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SaaS Billing & Revenue Management (BRM)

Modernized billing ingestion, revenue event processing, and operational reliability.

- Designed a Kafka-based ingestion pipeline using the outbox pattern, eliminating duplicate billing events and cutting escalations by **~80%**.
- Added deterministic idempotency layers ensuring consistent billing calculations across retries and crash scenarios.
- Rebuilt Airflow DAGs with schema validation, SLA monitoring, and late-arrival handling to stabilize nightly billing jobs.
- Implemented OpenTelemetry tracing across Java/Node/.NET billing components to unify debugging and dependency tracking.
- Created dashboards for billing spikes, ingestion lag, entitlement drift, and charge anomalies for finance and engineering teams.

Education

Cornell University — Master of Science, Electrical & Computer Engineering (2015–2017)

University of Liverpool — Bachelor of Science, Electrical & Electronic Engineering (2013–2015)

Xi'an Jiaotong-Liverpool University (XJTLU) — Bachelor of Science, Telecommunication Engineering (2011–2013)

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Amazon Web Services — Issued 2021

Microsoft Certified: Azure Developer Associate

Microsoft — Issued 2023

OpenAI LLM Application Developer Certificate

OpenAI — Completed 2024

KCNA: Kubernetes & Cloud Native Associate

The Linux Foundation — Issued 2022

Certified Scrum Developer (CSD)

Scrum Alliance — Issued 2019